

TITLE: Generalized Linear Models (GLMs)

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SUBJECT NAME: Machine Learning

SUBJECT CODE: PEC-CS701E

NAME OF COLLEGE: St. Thomas' College of Engineering and Technology

DATE: 31.07.25

INTRODUCTION

- **Definition (GLM):**
 - Extends linear regression models.
 - Allows response variable from **Exponential Family** (e.g., Normal, Binomial, Poisson, Gamma).
 - Connects linear predictors to mean via a **Link Function**.
- **Motivation:**
 - Handles different data types: **binary, count, skewed continuous**.
 - No need for **data transformation**; variance is modelled as a function of the mean.
 - Provides a **unified estimation** approach (e.g., **IRLS**) that fits multiple model types like logistic, Poisson, and linear.

COMPONENTS

- **Random Component:** Response variable Y follows a distribution from the **exponential family** (e.g., Binomial, Poisson, Gamma, Normal).
- **Systematic Component:** Uses a **linear predictor** $\eta = X\beta$, combining input variables X with coefficients β .
- **Link Function:**
 - Connects the **mean μ** of the response to the predictor via $g(\mu) = \eta$.
 - Examples: **logit, log, identity**.
 - Ensures output is on a valid scale (e.g., probability between 0 and 1).

COMMON DISTRIBUTION – LINK PAIRS

Distribution	Link Function	Example Application
Binomial/Bernoulli	Logit (log-odds)	Logistic regression for binary outcomes
Poisson	Log link	Modeling count data (e.g. accident rates)
Normal (Gaussian)	Identity link	Standard linear regression
Gamma	Log or inverse link	Modeling skewed positive outcomes

PARAMETER ESTIMATION

- **Maximum Likelihood Estimation (MLE):**
 - Estimates parameters β by **maximizing the likelihood** of observed data.
 - The likelihood is defined based on the **chosen distribution** (e.g., Binomial, Poisson).
 - MLE provides a **general approach** for fitting all GLMs.
- **Common Algorithms:**
 - **Iteratively Re-weighted Least Squares (IRLS):**
 - Efficient method for solving MLE in GLMs.
 - Equivalent to **Fisher Scoring** when using a **canonical link function**.
 - **Newton–Raphson / Fisher Scoring:**
 - Updates parameters β **iteratively**.
 - Continues until **convergence** is achieved (no significant change in estimates).

ADVANTAGES

- **Flexibility:** Handles many response types—binary, count, skewed continuous.
- **Interpretability:** Coefficients meaningfully describe predictor effects; clear hypothesis testing available.
- **Robustness:** Works under non-normality, accommodates heteroscedasticity; link and variance functions relax OLS assumptions.
- **Unified Framework:** One estimation process for various models (logistic, Poisson, linear regression) using the same software/function in R, SAS, etc.

MODEL DIAGNOSTICS

- **Deviance:** Compares model fit to saturated model; lower deviance = better fit.
- **Inference:** Use Wald, Likelihood-ratio, and Score tests for hypothesis testing of coefficients.
- **Residual checks** (e.g. deviance residuals) help detect overdispersion or model misspecification.
- **Quasi-likelihood models** handle cases where variance exceeds what the exponential family predicts

GEE and GLMM

- **Generalized Estimating Equations (GEE):**
 - A marginal modeling method for **correlated or repeated measures data** (e.g., longitudinal, clustered).
 - Estimates **population-averaged effects** without modeling subject-level random effects.
 - Robust to misspecification of the correlation structure; widely used in public health and social sciences.
- **Generalized Linear Mixed Models (GLMMs):**
 - Extends GLMs by **incorporating random effects**, allowing modeling of **subject-specific variability**.
 - Suitable for hierarchical or multi-level data structures (e.g., students within schools).
 - Estimation involves complex integration, often solved via **Laplace approximation** or **adaptive quadrature**.

CONCLUSION

- GLMs generalize regression modeling by allowing **non-normal outcomes** via flexible distributions and link functions.
- They unify logistic, Poisson, gamma, and normal regression under one framework, with consistent estimation techniques.
- GLMs reconcile mean–variance relationships naturally, eliminating the need for ad hoc transformations.
- Extensions like **GEE** and **GLMM** allow handling **correlated** and **hierarchical data** effectively.
- Overall, GLMs offer a powerful, interpretable, and statistically rigorous framework for modern predictive modeling and inference.

REFERENCES

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